

Website Resolution — Diagnostic Findings

Path B (SERP) / Path C (LLM) instrumentation · `enrichment_api` · branch `diag/website-trace` · live SerpAPI run

Headline: For two of the three failing companies, the company's own site **never appeared in the SERP result set at all** — a **retrieval** problem, not a guard rejecting it. The third (`Verdorex, Inc.`) **resolved correctly** in the live run. The SERP result sets have **drifted** since these records were characterized, so the "currently succeeds / fails" table no longer reproduces exactly.

1 · Setup confirmed before instrumenting

- Rejection points & real short-circuit order** in `select_website_from_serp`: `url_shape ( _URL_RE.match ) → blacklist ( _is_blacklisted ) → name_overlap ( _name_overlap )`, then ranking — `rank_0` = a Name 1 token is **not** in the host (`_name_in_domain` false), `1` = foreign-brand label, `2` = clean host match.
- Cache would not interfere:** the trace script uses a fresh, unshared `BatchCache` per record, so every query hits the wire.
- No behaviour change:** trace is a read-only re-evaluation of the same pure guards, gated behind `WEBSITE_TRACE` (default off). Full suite: **974 passed** (only 3 unrelated pre-existing failures); off = zero trace records.

2 · Outcome for all six records

Record	Results returned	Outcome	Why
Atlantic Testing Labs (US)	5	empty → Path C null	Own site never in results ; 4× <code>name_overlap</code> , 1× <code>blacklist</code>
Fine Organics Limited (GB)	5	empty → Path C null	Own site never in results ; 1× <code>blacklist</code> , 4× <code>rank_0</code> (matched <code>fine</code> @title only)
Verdorex, Inc. (US)	5	<code>https://www.verdorex.com</code> (rank 2, high)	Resolved correctly — position 1 clean host match
Silverline Biotech (US) [control]	1	empty → Path C null	Only 1 irrelevant result (<code>tealhq.com</code>) returned
Belharra Therapeutics (US) [control]	5	<code>belharratx.com</code> (rank 2, high)	Chose position 1 clean host
Suncoast Medical (US) [control]	5	<code>suncoastmg.com</code> (rank 2, high)	Chose position 1 clean host

3 · Per-candidate detail — the two retrieval misses

Atlantic Testing Labs — "Atlantic Testing Labs" official website US

`atlantictesting.com` is absent from the result set. Every returned host is genuinely unrelated and correctly dropped:

#	rejected_by	host
1	<code>name_overlap</code>	<code>nvlpubs.nist.gov</code>
2	<code>name_overlap</code>	<code>www.co.essex.ny.us</code>
3	<code>name_overlap</code>	<code>www.govinfo.gov</code>
4	<code>blacklist</code>	<code>www.facebook.com</code>
5	<code>name_overlap</code>	<code>steubencountyny.gov</code>

Fine Organics Limited — "Fine Organics Limited" official website GB

`fineorganics.com` is absent. The pages that returned are directory/registry entries where `fine` appears in the **title** only, so `rank_0` correctly refused them — the guard behaved exactly right:

#	rejected_by	rank	matched	host
1	blacklist	—	—	uk.linkedin.com
2	rank_0	0	fine @title	find-and-update.company-information.service.gov.uk
3	rank_0	0	fine @title	www.gov.uk
4	rank_0	0	fine @title	www.cphi-online.com
5	rank_0	0	fine @title	www.pharmacompass.com

In both cases Path C (LLM) also returned no URL, so nothing was written.

4 · Read of why they failed

- **Atlantic Testing Labs** — no guard touched the real site because the real site was never a candidate. Pure retrieval miss.
- **Fine Organics Limited** — the homepage wasn't retrieved; the guard correctly rejected the directory pages that were. Retrieval miss + a correctly-firing guard.
- **Verdox, Inc.** — **did not fail** in the live run; resolved to `verdox.com` at rank 2 / high.

SERP drift. The result sets have changed since these records were characterized, so the controls no longer match the earlier table either: Silverline returned a single irrelevant result and fell through; Suncoast now picks `suncoastmg.com` (a different valid rank-2 host); Belharra resolved to `belharratx.com` rather than the listed `carricktherapeutics.com`. "Verdox fails / these controls succeed" reflects an **earlier SERP snapshot**, not current behaviour.

Hypotheses for the retrieval miss (noted, not applied — per the hard constraint)

- Exact-phrase quoting `"Atlantic Testing Labs"` can exclude a site that brands itself slightly differently ("...Laboratories").
- Country-only geo (no city/state) lets directory/registry pages outrank the homepage for these names.
- `num_results=5` is shallow — a real homepage at position 6+ is never seen.
- Path C didn't rescue them because the LLM also returned no URL.

None of these are guard bugs; the guards did their job on the candidates they were given.

Instrumentation: `WEBSITE_TRACE` flag (default off), read-only per-candidate JSON trace on the `enrichment.trace.website` logger, and `scripts/trace_website.py`. Raw trace: `logs/website_trace.json` (9 JSON lines).